CU CUET Prep

Gravitation

Why does an apple fall towards the ground?

Physics · Learning notes with a mind map and a flowchart · Written from NCERT, Physics Part I, Textbook for Class XI, Chapter 7: Gravitation (Reprint 2026-27)

Preview: not yet reviewed. These notes have not yet been checked by a subject expert. Gravitation is a Class 11 topic and is not in the CUET (UG) Physics syllabus; it is used here to try out this way of learning.

1. Bodies fall towards the earth

All material objects are attracted towards the earth. Anything thrown up falls down, raindrops fall from the clouds, and going uphill is more tiring than going downhill.

Galileo recognised that all bodies, irrespective of their masses, are accelerated towards the earth with a constant acceleration.

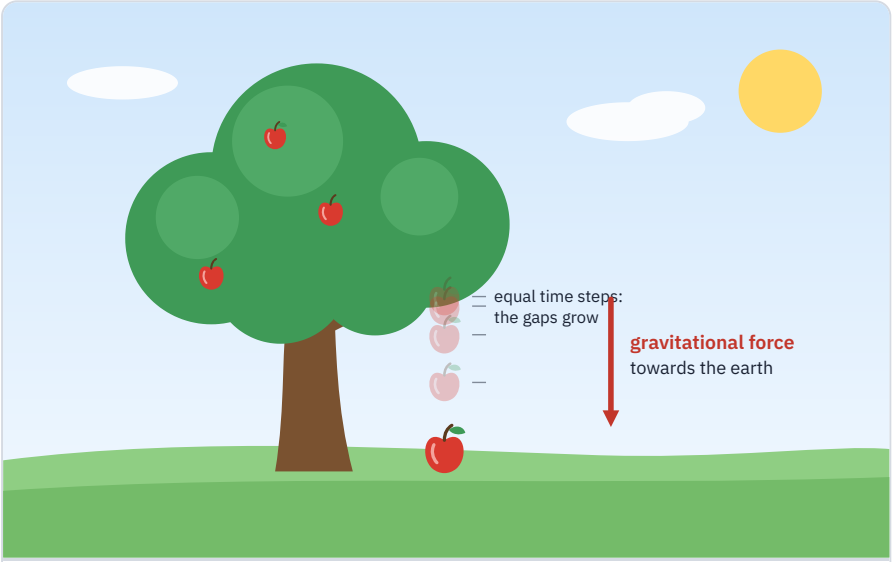


Figure: the apple falls straight down. In equal time steps it covers more and more distance, because it accelerates.

Source: NCERT Class XI Physics, section 7.1, page 127

2. The gravitational force

The earth attracts the apple, and the apple attracts the earth. The two forces are equal in size and opposite in direction (Newton's third law): $F_{12} = -F_{21}$.

- The force is attractive.
- It acts along the line joining the two bodies.
- A sphere whose mass is spread symmetrically attracts an outside body as though all its mass were at its centre.

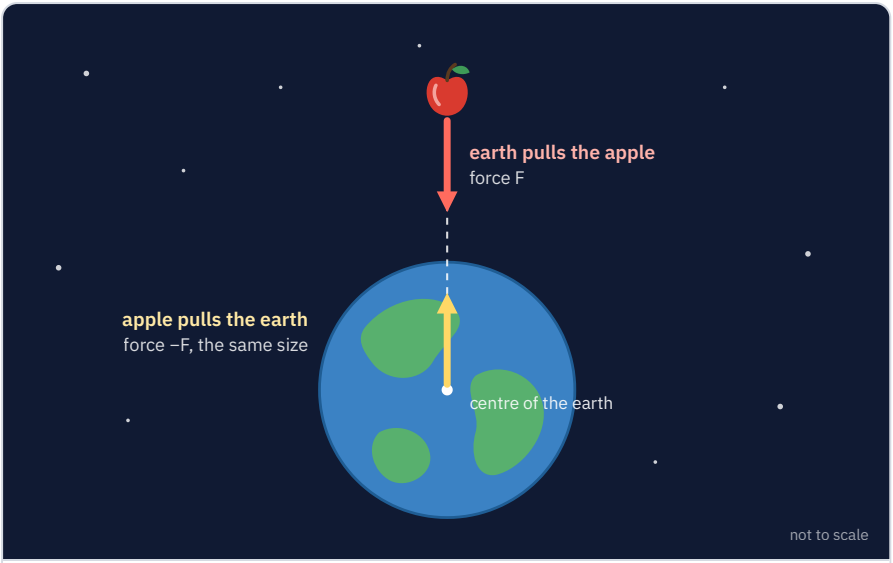


Figure: the earth pulls the apple and the apple pulls the earth with forces of the same size, along the line joining them. Not to scale.

Source: NCERT Class XI Physics, section 7.3, page 130

3. Newton’s universal law of gravitation

Every body in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

The law is stated for point masses. When there are several masses, each force acts independently and the resultant is found by vector addition (the principle of superposition).

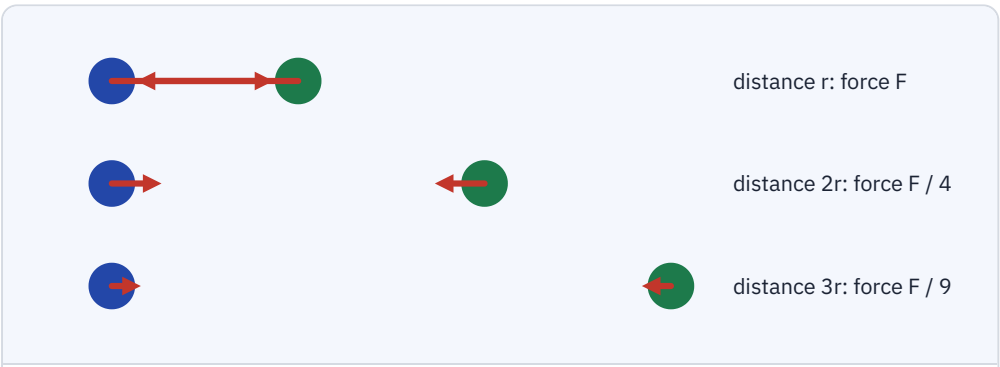


Figure: the force falls as the square of the distance.

Universal law of gravitation

$$F = G\,\frac{m_1 m_2}{r^2}$$

F	gravitational force between the two bodies (newton, N)
G	universal gravitational constant, 6.67×10^{-11} N m^2 kg^{-2}
$m_1,\ m_2$	masses of the two bodies (kg)
r	distance between them (m)

Example from the chapter: the apple and the moon

The moon's centripetal acceleration is $a_m = 4\pi^2 R_m / T^2$. With T ≈ 27.3 days and R_m ≈ 3.84 × 10^8 m this is about 0.0027 m s^{-2}, much smaller than g ≈ 9.8 m s^{-2}. If the force falls as the inverse square of the distance, g/a_m should equal R_m^2/R_E^2 ≈ 3600, and it does.

Source: NCERT Class XI Physics, section 7.3, page 130

4. The gravitational constant G

G is the universal gravitational constant. Its value is 6.67 × 10^{-11} N m^2 kg^{-2}. It was first measured in Cavendish's experiment.

Source: NCERT Class XI Physics, section 7.4, page 139

5. Acceleration due to gravity

At the surface of the earth the gravitational force on a body of mass m is $F = GM_E m / R_E^2$. The acceleration it produces is called g, and F = mg.

g is easy to measure and R_E is known. Together with G from Cavendish's experiment they give the mass of the earth. Hence the saying "Cavendish weighed the earth".

Acceleration due to gravity at the earth's surface

$$g = \frac{F}{m} = \frac{GM_E}{R_E^2}$$

g	acceleration due to gravity, about 9.8 m s^{-2}
M_E	mass of the earth
R_E	radius of the earth
$F = mg$	force on a body of mass m at the surface

Source: NCERT Class XI Physics, section 7.5, page 133

6. g above and below the surface

At a height h the body is at a distance $R_E + h$ from the centre of the earth, so g is smaller than at the surface.

At a depth d only the sphere of radius $R_E - d$ below the body attracts it; the outer shell exerts no force on it. So g is smaller there too: $g(d) = g(1 - d/R_E)$.

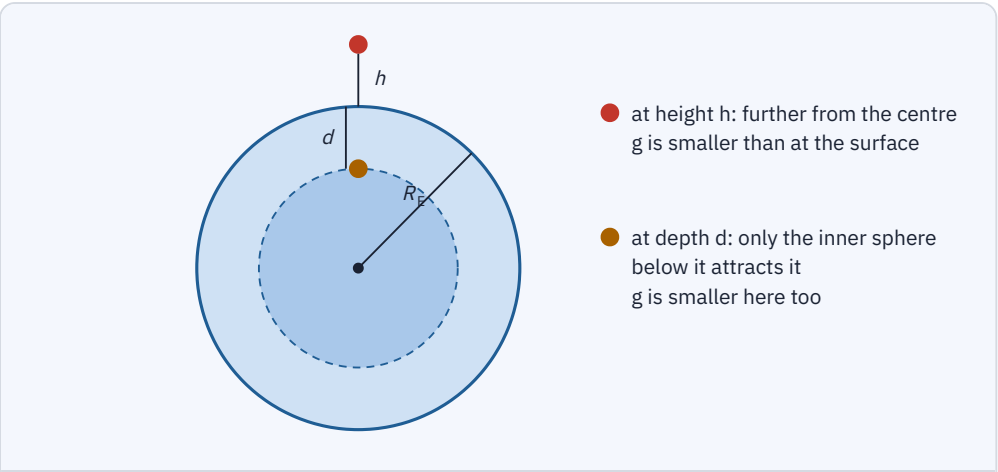


Figure: a body above the surface and a body below it.

At a height h above the surface

$$g(h) = \frac{GM_E}{(R_E + h)^2} \approx g\left(1 - \frac{2h}{R_E}\right) \quad (h \ll R_E)$$

h	height above the earth's surface
$g(h)$	acceleration due to gravity at that height: less than g

At a depth d below the surface

$$g(d) = g\left(1 - \frac{d}{R_E}\right)$$

d	depth below the earth's surface
$g(d)$	acceleration due to gravity at that depth: also less than g

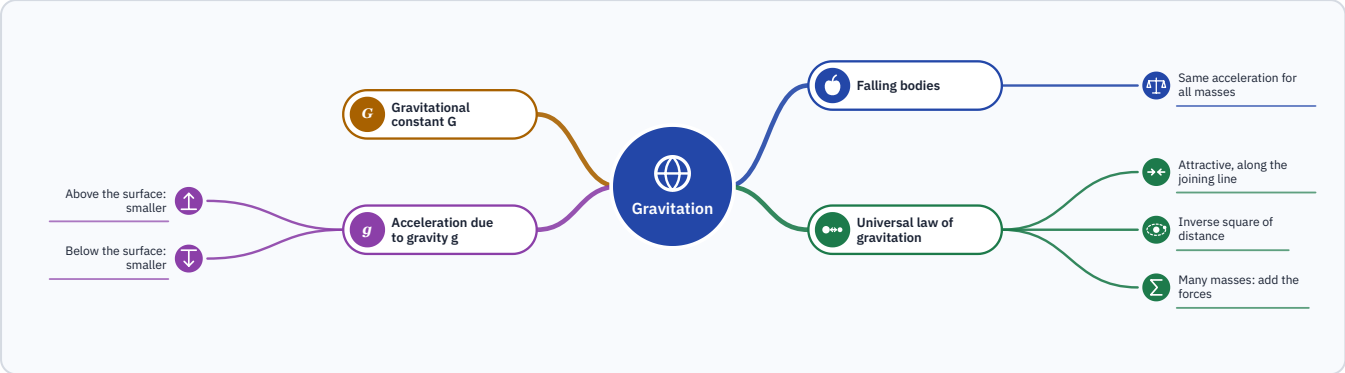
Source: NCERT Class XI Physics, section 7.6, page 133

Points to remember

- The gravitational force is always attractive and acts along the line joining the two bodies.
- F = G $m_1 m_2$ / r^2. Double a mass: the force doubles. Double the distance: the force becomes one quarter.
- G = 6.67 × 10^{-11} N m^2 kg^{-2} is the same everywhere; g ≈ 9.8 m s^{-2} is its effect at the earth's surface.
- $g = GM_E/R_E^2$ does not depend on the mass of the falling body.
- g decreases both above the surface (height h) and below it (depth d).

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Mind map: Gravitation



Each branch is one idea from the notes. Follow a branch outwards for its details. The ideas are listed below.

Falling bodies
All material objects are attracted towards the earth. Anything thrown up falls down.

> **Same acceleration for all masses**
> Galileo recognised that all bodies, irrespective of their masses, are accelerated towards the earth with a constant acceleration.

Universal law of gravitation
Every body attracts every other body with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

> **Attractive, along the joining line**
> The gravitational force is attractive. The forces on the two bodies are equal in size and opposite in direction: $F_{12} = -F_{21}$.

> **Inverse square of distance**
> Twice the distance gives one quarter of the force. Newton compared the moon's acceleration with g: the ratio g/a_m equals R_m^2/R_E^2, about 3600.

Many masses: add the forces
The law is stated for point masses. For several masses, each force acts independently, and the resultant is found by vector addition (the principle of superposition).

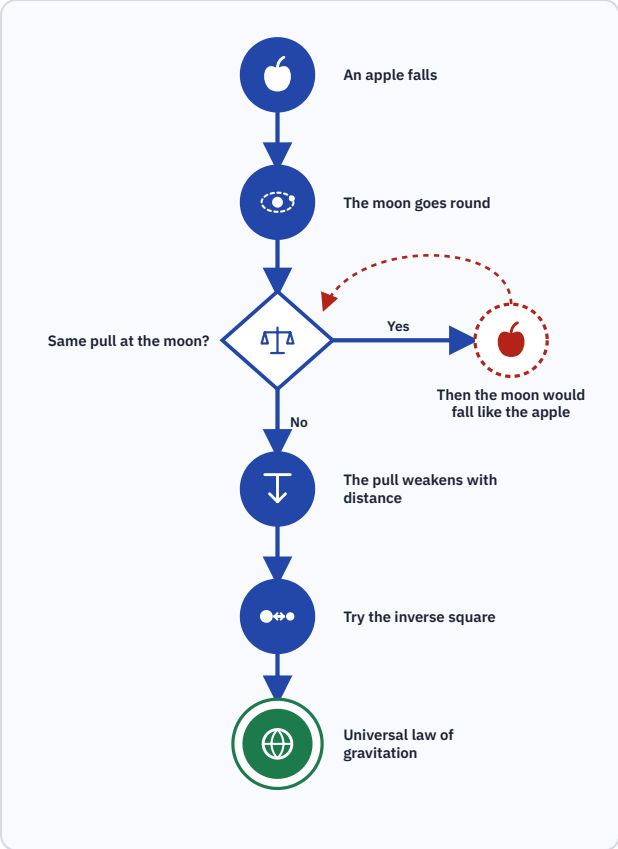
Gravitational constant G
G is the universal gravitational constant. Its value is 6.67×10^{-11} N m² kg⁻². It was first measured by Cavendish's experiment.

Acceleration due to gravity g
At the earth's surface the force on a mass m is F = mg, so $g = GM_E/R_E^2$. Its value is about 9.8 m s⁻². Knowing g, G and R_E gives the mass of the earth: "Cavendish weighed the earth".

> **Above the surface: smaller**
> At a height h the distance from the earth's centre is $R_E + h$, so g is less. For small heights it falls by the factor $(1 - 2h/R_E)$.

> **Below the surface: smaller**
> At a depth d only the sphere of radius $R_E - d$ below the point pulls on it; the shell outside exerts no force. So g falls by the factor $(1 - d/R_E)$.

Flowchart: from the falling apple to the universal law



1. **Stage 1: an apple falls.** Legend has it that seeing an apple fall from a tree set Newton thinking. Near the earth's surface every body falls with the acceleration due to gravity, g ≈ 9.8 m s^{-2}.
2. **Stage 2: the moon goes round the earth.** The moon moves in an orbit of radius R_m, so it has a centripetal acceleration towards the earth. Newton reasoned that this, too, is due to the earth's gravity. With T ≈ 27.3 days and R_m ≈ 3.84 × 10^8 m, the formula gives about 0.0027 m s^{-2}.
3. **Decision: is the earth's pull the same at the moon's distance?.** The apple's acceleration is 9.8 m s^{-2}. The moon's is about 0.0027 m s^{-2}. Is the earth's pull the same at both distances? No: the moon's acceleration is much smaller.
4. **Stage 3: the force due to earth's gravity decreases with distance.** The moon's acceleration is much smaller than g. This clearly shows that the force due to the earth's gravity decreases with distance. The question is: how fast?
5. **Stage 4: assume the inverse square of the distance.** If the force falls as the inverse square of the distance from the centre of the earth, then g/a_m should equal R_m^2/R_E^2, which is about 3600. Measured values agree: 9.8 ÷ 0.0027 is about 3600.
6. **Result: the universal law of gravitation.** These observations led Newton to propose the law: every body in the universe attracts every other body with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

Check yourself

1. What causes the apple to fall towards Earth?

A. The gravitational force of attraction between the earth and the apple

B. The air pushing the apple down

C. The apple is heavy, and only heavy things fall

D. Only the earth pulls; the apple exerts no force

2. The distance between two bodies is made three times as large. The gravitational force between them becomes:

A. one ninth

B. one third

C. three times

D. unchanged

Answers

1. (A) Every body attracts every other body. The earth's pull on the apple is the gravitational force, and it points towards the centre of the earth.

2. (A) The force is inversely proportional to the square of the distance: $3^2 = 9$, so the force is one ninth.